Null space

Solution space of homogeneous system.

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \quad \text{Null}(A) = ?$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 4 & 0 \end{array}\right) \xrightarrow{R_2' = R_2 - 2R_1} \left(\begin{array}{cc|c} \overset{x}{1} & \overset{y}{2} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$y = t, \quad x = -2y = -2t$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2t \\ t \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} t$$

$$\text{Null}(A) = \underbrace{\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} t \;\middle|\; t \in \mathbb{R} \right\}}_{\text{set}}$$

Null(A) is a subspace of $\mathbb{R}^n$.

① $\text{Null}(A) \neq \emptyset$ $\qquad \vec{0} \in \text{Null}(A)$

② $u, v \in \text{Null}(A)$ $\qquad (u+v) \in \text{Null}(A)$

$Au = 0$

$Av = 0$

$$A(u+v) = Au + Av$$

$$= 0 + 0 = 0$$

③ $u \in \text{Null}(A)$ $\qquad \alpha u \in \text{Null}(A)$

$Au = 0$

$$A(\alpha u) = \alpha (Au)$$

$$= \alpha \times 0 = 0.$$

Example

$Tx = 0$

$$\Rightarrow \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 7 & -4 & 0 \end{array}\right) \underset{R_3' = 2R_3 - R_1}{\overset{R_2' = 2R_2 - R_1}{\sim}} \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 15 & -9 & 0 \end{array}\right)$$

$R_3' = R_3 - 3R_2$

$$\sim \begin{array}{c} \begin{array}{cccc} \alpha & \beta & \gamma & \end{array} \\ \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

$\gamma = t$, $\quad 5\beta - 3\gamma = 0 \Rightarrow 5\beta = 3t$

$$\beta = \frac{3}{5}t$$

$$2\alpha - \beta + \gamma = 0 \Rightarrow 2\alpha = \frac{3}{5}t - t$$

$$= \left(\frac{3-5}{5}\right)t$$

$$\cancel{2}\alpha = -\frac{\cancel{2}}{5}t$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix} t$$

$$\text{Null}(T) = \left\{ \begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix} t \;\middle|\; t \in \mathbb{R} \right\}$$

V_1

Dimension of Null Space of T = $\square$

Nullity

<u>Definition</u>: Given, A

$$\underline{\text{Null}(A)} = \{ x \mid \underline{Ax = 0} \}$$

solution of this system

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 7 \end{pmatrix}$$

Zero vector

$$\begin{array}{cccc} \alpha & \beta & \gamma & \delta \end{array}$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 7 \end{pmatrix}_{2\times 4} \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}_{4\times 1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}_{2\times 1}$$

Pivot α β Pivot γ δ

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 0 \\ & & & & \end{pmatrix}$$

$\beta = t$ $\delta = s$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix} \qquad \beta = t \quad \delta = s$$

New Example:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : \begin{cases} x - y = 0 \\ y + z = 0 \end{cases} \right\}$$

$$\begin{aligned} x - y + 0z &= 0 \\ 0x + y + z &= 0 \end{aligned}$$

$$\underbrace{\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Example (Row Space)

$$T = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix}$$

$$\underbrace{\mathrm{Row}(T)} = \alpha(2,-1,1) + \beta(1,2,-1) + \gamma(1,7,-4)$$

$$= \underbrace{(2\alpha+\beta+\gamma,\ -\alpha+2\beta+7\gamma,\ \alpha-\beta-4\gamma)}$$

$$= \left\{ (2\alpha+\beta+\gamma,\ -\alpha+2\beta+7\gamma,\ \alpha-\beta-4\gamma) \,\middle|\, \alpha,\beta,\gamma \in \mathbb{R} \right\}$$

<u>Motivation:</u> $\mathbb{R}^2$, $\left\{ \underbrace{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}, \cancel{\begin{pmatrix}2\\2\end{pmatrix}}} \right\}$

basis $= \left\{ \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix} \right\}$

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix} \sim\sim\sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 5 & -3 \\ 0 & 0 & \boxed{0} \end{pmatrix}$$

this equation can be generated by other.

$$\underbrace{\text{Row}(A)} = \alpha(2,-1,1) + \beta(0,5,-3)$$

$$= (2\alpha, -\alpha+5\beta, \alpha-3\beta)$$

Find the Basis of Row(A)?

$$B = \{\underbrace{(2,-1,1)}_{v_1}, \underbrace{(1,2,-1)}_{v_2}\}$$

Here we need $\underbrace{\text{two vectors}}_{v_1, v_2}$ to generate row space

$$\boxed{\text{dimension of Row space}} = 2$$

$= \text{Row Rank}$

<u>N.B.:</u> Row operation doesn't change the Row span)

<u>Example</u> (Column space)

Example (Column space)

$$T = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix}$$

$$Col(T) = \alpha(2,1,1) + \beta(-1,2,7) + \gamma(1,-1,-4)$$

(Method 1)

$$\begin{pmatrix} 2 & 1 & 1 \\ -1 & 2 & 7 \\ 1 & -1 & -4 \end{pmatrix} \xrightarrow{R_2' = 2R_2 + R_1} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 5 & 15 \\ 0 & -3 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -9 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$cs(A) = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$$

Method 2

pivot pivot

independent

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix} \sim \sim \sim \begin{pmatrix} 2 & -1 & 1 \\ 0 & 5 & -3 \\ 0 & 0 & \boxed{0} \end{pmatrix}$$

original

REF

$$= \text{Span}\{(2,1,1), (-1,2,7)\}$$

column space $= \{\alpha(2,1,1) + \beta(-1,2,7)\}$

$$= (2\alpha - \beta, \alpha + 2\beta, \alpha + 7\beta)$$

or $\begin{pmatrix} 2\alpha - \beta \\ \alpha + 2\beta \\ \alpha + 7\beta \end{pmatrix}$

basis $= \{(2,1,1), (-1,2,7)\}$

$\langle \alpha, \beta \rangle = \{ (2,1,1), (-1,2,1) \}$

N.B: (Row operation change the column space)